Bioinformatics algorithms

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Sequence alignment – exercises (**with solutions**)

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1. **Global sequence alignment 1**

   Compute the global sequence alignment of the following two strings "DESPERATE" and "ASPARAGUS", using these score parameters:

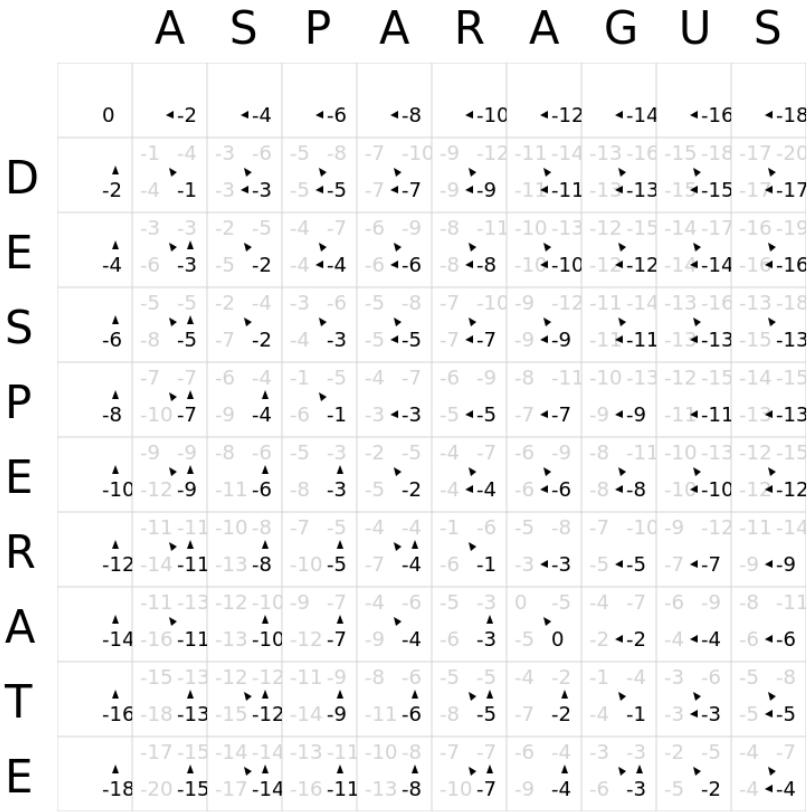
   - Gap penalty: -2
   - Mismatch penalty: -1
   - Match score: 1

   Answer the following questions:

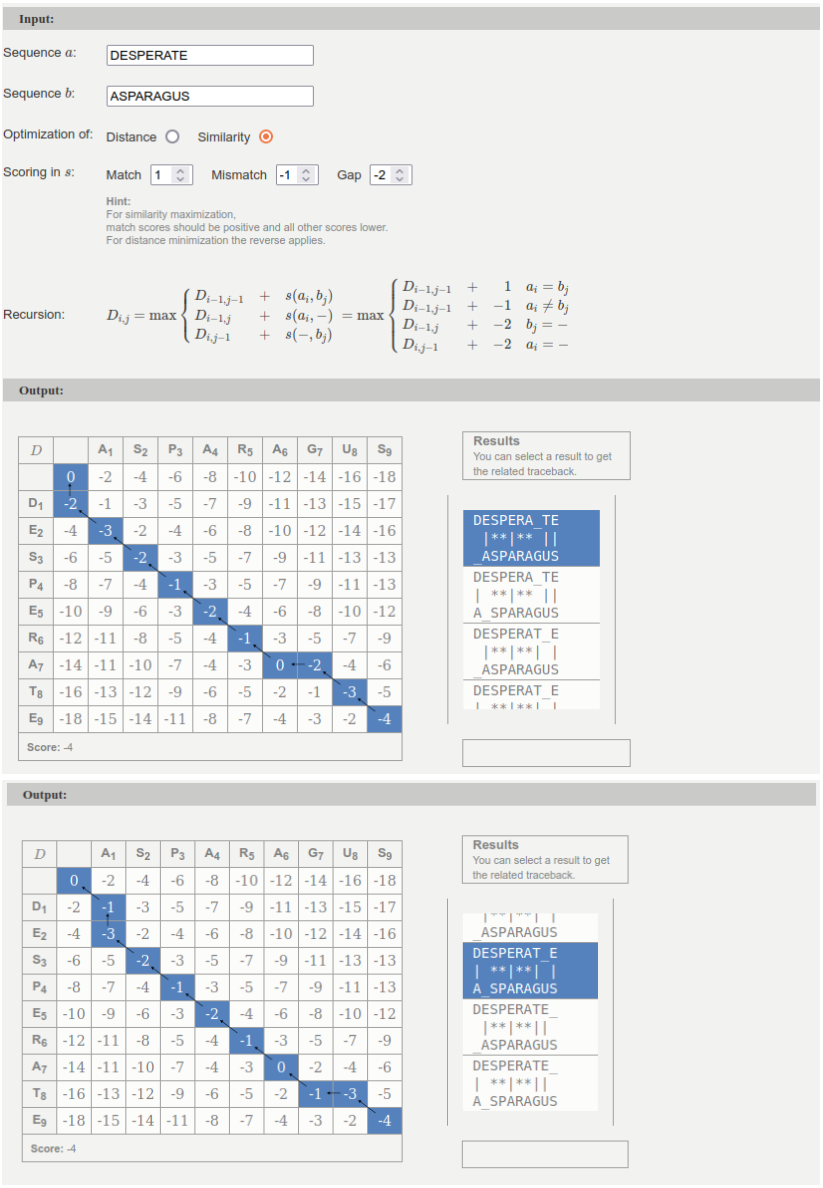
   (a) What is the final alignment score?

   (b) Are there multiple equivalent solutions?

   Solution



(from https://blievrouw.github.io/needleman-wunsch/)



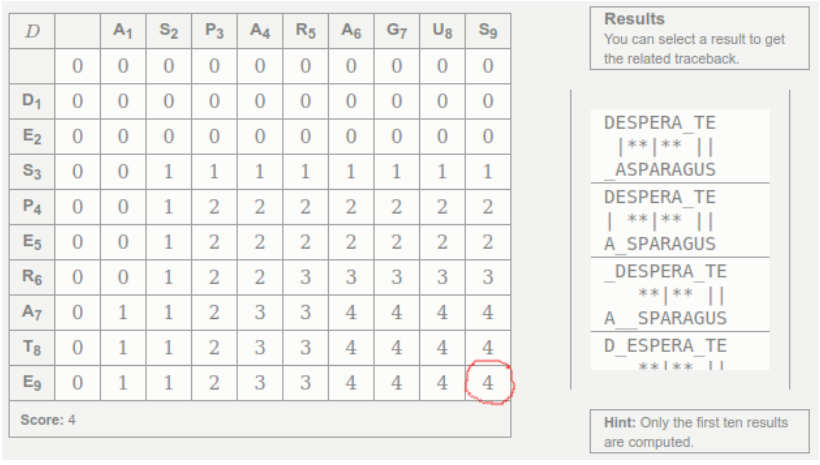
(from `http://rna.informatik.uni-freiburg.de/Teaching/index.jsp?toolName=Needleman-Wunsch`)

2. **Longest common subsequence (LCS)**

   Determine the length of the longest common subsequence of the two strings from the first exercise. You can modify the previous exercise in a very simple way to find the answer ...

   Solution

   We can simply set the following parameters: match=1, mismatch=0, gap=0



   Result: 4 (`SPRA`)

3. **Global sequence alignment 2**

Compute the global sequence alignment of the following two DNA sequences "AGCTTAGTTA" and "CGTTAGTAG", using these score parameters:

- Gap penalty: -2
- Mismatch penalty: -1
- Match score: 1

Answer the following questions:

(a) What is the final alignment score?

(b) Are there multiple equivalent solutions?

Solution

		C	G	T	T	A	G	T	A	G
	0	◂-2	◂-4	◂-6	◂-8	◂-10	◂-12	◂-14	◂-16	◂-18
A	▴-2	-1 -4 ▸-1	-3 -6 ▸◂-3	-5 -8 ▸◂-5	-7 -10 ▸◂-7	-7 -12 ▸-7	-11 -14 ◂-9	-13 -16 ◂-11	-13 -18 ▸◂-13	-17 -20 ◂-15
G	▴-4	-3 -3 ▸▴-3	0 -5 ▸0	-4 -7 ◂-2	-6 -9 ◂-4	-8 -9 ◂-6	-6 -11 ▸-6	-10 -13 ◂-8	-12 -15 ◂-10	-12 -17 ▸◂-12
C	▴-6	-3 -5 ▸-3	-4 -2 ▴-2	-1 -4 ▸-1	-3 -6 ▸◂-3	-5 -8 ▸◂-5	-7 -8 ▸◂-7	-7 -10 ▸-7	-9 -12 ▸◂-9	-11 -14 ▸◂-11
T	▴-8	-7 -5 ▴-5	-4 -4 ▸▴-4	-1 -3 ▸-1	0 -5 ▸0	-4 -7 ◂-2	-6 -9 ◂-4	-6 -9 ▸◂-6	-8 -11 ◂-8	-10 -13 ▸◂-10
T	▴-10	-9 -7 ▴-7	-6 -6 ▸▴-6	-3 -3 ▸▴-3	0 -2 ▸0	-1 -4 ▸-1	-3 -6 ▸◂-3	-3 -8 ▸-3	-7 -10 ◂-5	-9 -12 ◂-7
A	▴-12	-11 -9 ▴-9	-8 -8 ▸▴-8	-7 -5 ▴-5	-4 -2 ▴-2	1 -3 ▸1	-2 -5 ◂-1	-4 -5 ◂-3	-2 -7 ▸-2	-6 -9 ◂-4
G	▴-14	-13 -11 ▴-11	-8 -10 ▸-8	-9 -7 ▴-7	-6 -4 ▴-4	-3 -1 ▴-1	2 -3 ▸2	-2 -5 ◂0	-4 -4 ◂-2	-1 -6 ▸-1
T	▴-16	-15 -13 ▴-13	-12 -10 ▴-10	-7 -9 ▸-7	-6 -6 ▸▴-6	-5 -3 ▴-3	-2 0 ▴0	3 -2 ▸3	-1 -4 1 ◂1	-3 -3 ◂-1
T	▴-18	-17 -15 ▴-15	-14 -12 ▴-12	-9 -9 ▸▴-9	-6 -8 ▸-6	-7 -5 ▴-5	-4 -2 ▴-2	1 1 ▸▴1	2 -1 ▸2	0 -3 ▸0
A	▴-20	-19 -17 ▴-17	-16 -14 ▴-14	-13 -11 ▴-11	-10 -8 ▴-8	-5 -7 ▸-5	-6 -4 ▴-4	-3 -1 ▴-1	2 0 ▸2	1 -2 ▸1

D		C_1	G_2	T_3	T_4	A_5	G_6	T_7	A_8	G_9
	0	-2	-4	-6	-8	-10	-12	-14	-16	-18
A_1	-2	-1	-3	-5	-7	-7	-9	-11	-13	-15
G_2	-4	-3	0	-2	-4	-6	-6	-8	-10	-12
C_3	-6	-3	-2	-1	-3	-5	-7	-7	-9	-11
T_4	-8	-5	-4	-1	0	-2	-4	-6	-8	-10
T_5	-10	-7	-6	-3	0	-1	-3	-3	-5	-7
A_6	-12	-9	-8	-5	-2	1	-1	-3	-2	-4
G_7	-14	-11	-8	-7	-4	-1	2	0	-2	-1
T_8	-16	-13	-10	-7	-6	-3	0	3	1	-1
T_9	-18	-15	-12	-9	-6	-5	-2	1	2	0
A_{10}	-20	-17	-14	-11	-8	-5	-4	-1	2	1

Score: 1

Results

You can select a result to get the related traceback.

```
AGCTTAGTTA
|* *****||
CG_TTAGTAG
```

(c) Verify that the final score you calculated is correct by inferring it also directly from the alignment which you obtained (helps to cross-check and identify potential errors in the procedure)!

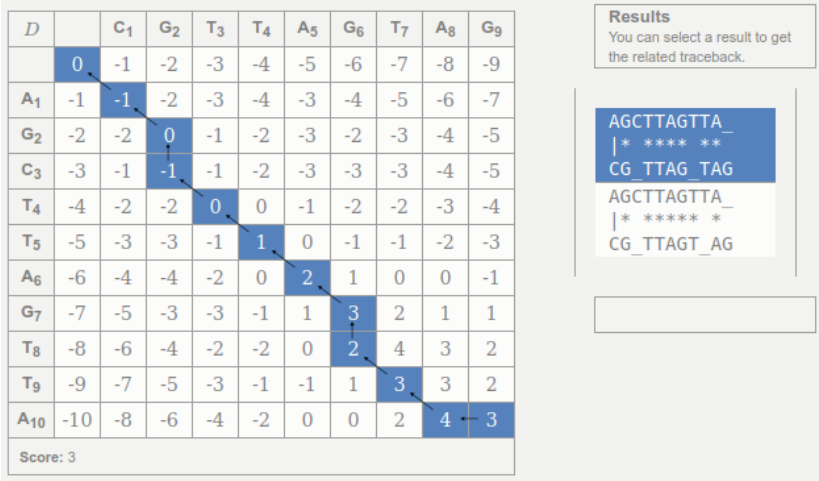
Solution

```
 A   G   C   T   T   A   G   T   T   A
 C   G   -   T   T   A   G   T   A   G

-1  +1  -2  +1  +1  +1  +1  +1  -1  -1  =  1
```

(d) Does it make any difference if you use a gap penalty of -1 instead (same as mismatch penalty)? Compute the alignment again and compare the results. Can you explain the differences you observe (if any)?

Solution



D		C_1	G_2	T_3	T_4	A_5	G_6	T_7	A_8	G_9
	0	-1	-2	-3	-4	-5	-6	-7	-8	-9
A_1	-1	-1	-2	-3	-4	-3	-4	-5	-6	-7
G_2	-2	-2	0	-1	-2	-3	-2	-3	-4	-5
C_3	-3	-1	-1	-1	-2	-3	-3	-3	-4	-5
T_4	-4	-2	-2	0	0	-1	-2	-2	-3	-4
T_5	-5	-3	-3	-1	1	0	-1	-1	-2	-3
A_6	-6	-4	-4	-2	0	2	1	0	0	-1
G_7	-7	-5	-3	-3	-1	1	3	2	1	1
T_8	-8	-6	-4	-2	-2	0	2	4	3	2
T_9	-9	-7	-5	-3	-1	-1	1	3	3	2
A_{10}	-10	-8	-6	-4	-2	0	0	2	4	3

Score: 3

Gaps are more tolerated now (it's less "costly" to insert a gap). The algorithm can insert two additional gaps, one at the end of the first sequence and one before the end "TAG" or before the end "AG" of the second sequence, and in turn it can align more identical nucleotides (7 here, for the gap penalty of -2 only 6 identical nucleotides could be aligned).

This means that **it is important how the parameters are set**! They should reflect the biological constraints. Are gaps (insertions and deletions) really as easy to obtain as mismatches (simple nucleotide changes/substitutions)? Usually the gaps are more penalized than simple mismatches!

4. ***Modified* global sequence alignment**

How could you modify the global sequence alignment, such that gaps (insertions and deletions) at the beginning and the end of the alignment have no penalty (gap penalty of 0 only at the beginning and end of the alignment)?

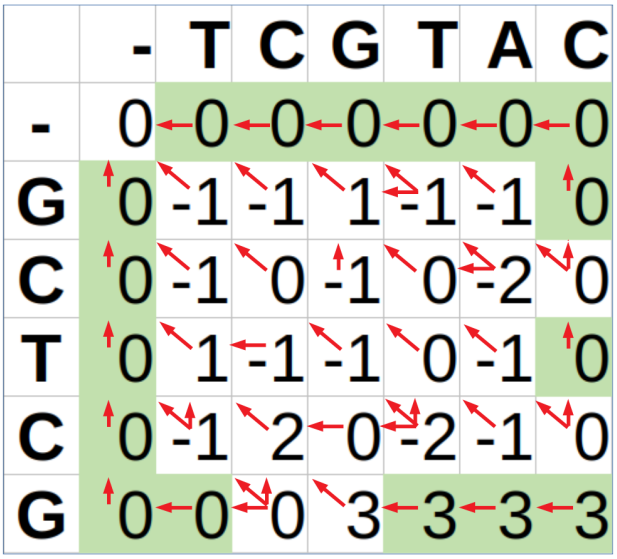
Apply your proposed approach to the following problem:

Find the best global alignment for the two sequences "TCGTAC" and "GCTCG" such that,

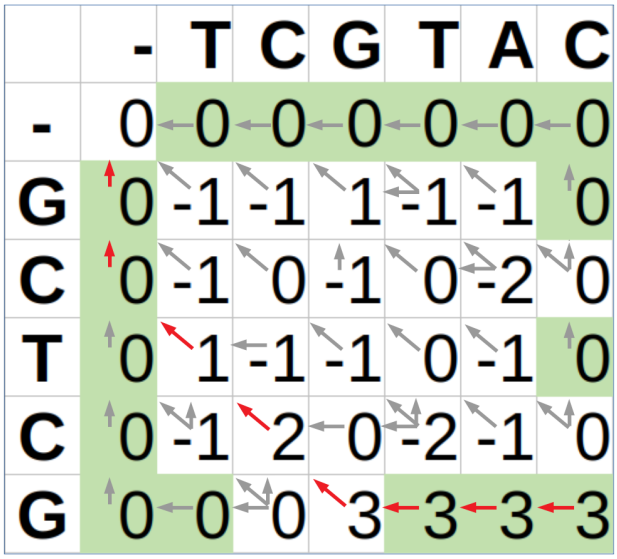
- match score = 1
- mismatch penalty = -1
- gap penalty = -2 (within the alignment)
- gap penalty = 0 (at the beginning and end of the alignment)

No need to write any formula, just apply your idea to the problem while you fill the scoring matrix!

Solution



	-	T	C	G	T	A	C
-	0	0	0	0	0	0	0
G	0	-1	-1	1	-1	-1	0
C	0	-1	0	-1	0	-2	0
T	0	1	-1	-1	0	-1	0
C	0	-1	2	0	-2	-1	0
G	0	0	0	3	3	3	3



	-	T	C	G	T	A	C
-	0	0	0	0	0	0	0
G	0	-1	-1	1	-1	-1	0
C	0	-1	0	-1	0	-2	0
T	0	1	-1	-1	0	-1	0
C	0	-1	2	0	-2	-1	0
G	0	0	0	3	3	3	3

The best global alignment is:

-	-	T	C	G	T	A	C
G	C	T	C	G	-	-	-